Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H1 BIOLOGY 8876/02

Paper 2 Structured Questions

14 September 2018

2 hours

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Answer **all** questions.

Write your answers on space provided in the Question Paper.

INFORMATION FOR CANDIDATES

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [ ].

For Examiner's Use	
Section A	
1	**/ 10**
2	**/ 5**
3	**/ 7**
4	**/ 8**
5	**/ 6**
6	**/ 9**
Section B	
1 / 2	**/ 15**
Total [60]	

Section A: Structured Questions (45 marks)

Answer **all** questions in this section.

For Examiner's use

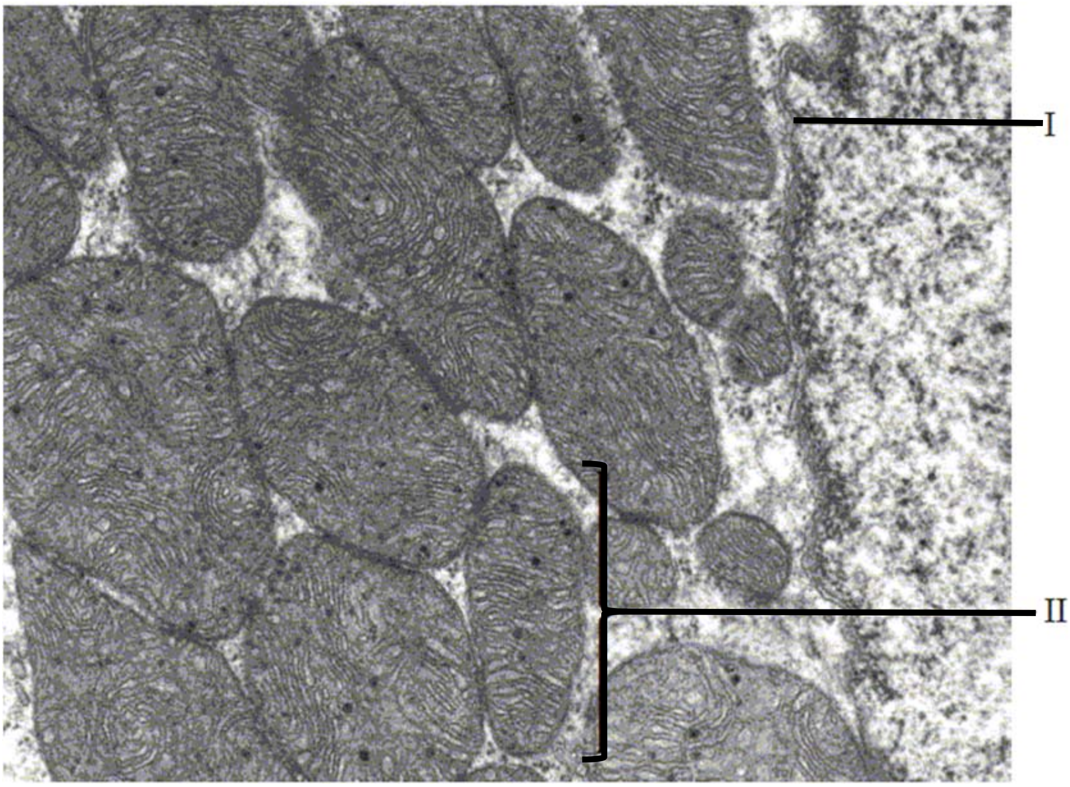
Question 1

(a) Distinguish between the terms "***resolution***" and "***magnification***". [1]

....

....

The electron micrograph below shows part of a cell.



(b) **(i)** Identify the structures labelled **I** and **II**. [2]

I : ....

II : ....

(ii) State one function of structure **I**. [1]

....

....

For Examiner's use

(iii) State one function of structure **II**. [1]

(iv) Deduce, with a reason, whether this cell is eukaryotic or prokaryotic. [1]

(c) RNAs are synthesised in nucleus and transported out into the cytoplasm for use in protein synthesis. Describe the roles of mRNA and rRNA in protein synthesis. [4]

mRNA:

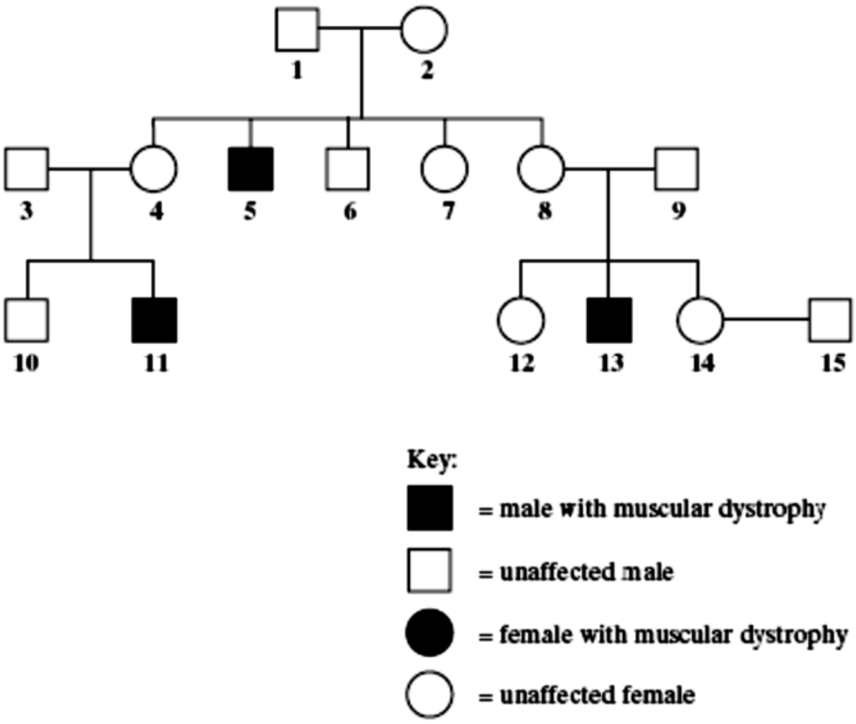
rRNA:

Total: [10]

Question 2

For Examiner's use

Duchenne muscular dystrophy is a sex-linked inherited condition which causes degeneration of muscle tissue. It is caused by a recessive allele. The diagram shows the inheritance of muscular dystrophy in one family.



(a) Give evidence from the diagram which suggests that muscular dystrophy is:

(i) sex-linked. [1]

....

....

(ii) caused by a recessive allele. [1]

....

....

For Examiner's use

(b) Using the following symbols:

X^D= an X chromosome carrying the normal allele
X^d= an X chromosome carrying the allele for muscular dystrophy
Y = a Y chromosome

Give **ALL** the possible genotypes of each of the following persons. [2]

5:

6:

7:

8:

(c) A blood test shows that person **14** is a carrier of muscular dystrophy. Person **15** has recently married person **14** but as yet they have had no children.

What is the probability that their first child will be a male who develops muscular dystrophy? [1]

Total: [5]

For Examiner's use

Question 3

(a) The *APP* gene provides instructions for making a protein called amyloid precursor protein. This protein is found in many tissues and organs, including the brain and spinal cord. The most common mutation on the *APP* gene involves one codon being changed from GCC to GUC. Fig. 3.1 shows the genetic code.

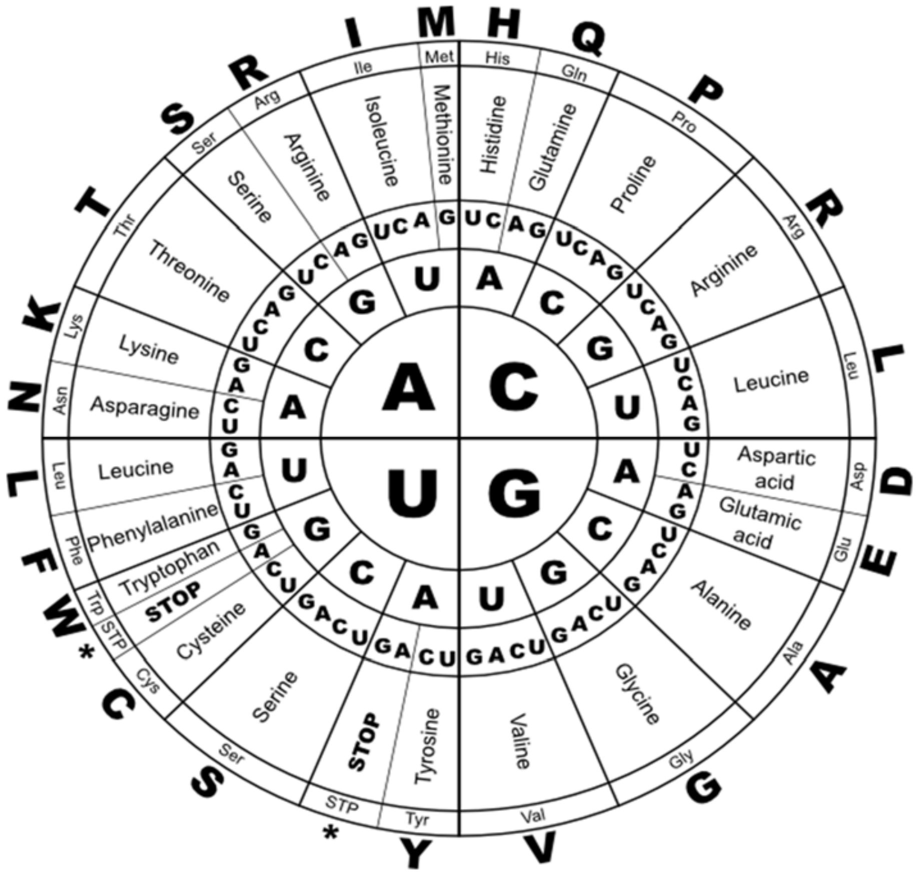


Fig. 3.1

Explain the significance of codon in translation. [3]

....

....

....

....

....

For Examiner's use

(b) With reference to Fig. 3.1, describe the mutation that has occurred. [2]

(c) Amyloid precursor protein is cut by enzymes to create smaller fragments, some of which are released outside the cell. Two of these fragments are called soluble amyloid precursor protein (sAPP) and amyloid beta (β) peptide. This mutation in the *APP* gene can lead to the production of a "stickier" form of the β peptide. When these protein fragments are released from the cell, they can accumulate in the brain and form clumps called amyloid plaques. These plaques are characteristic of Alzheimer disease.

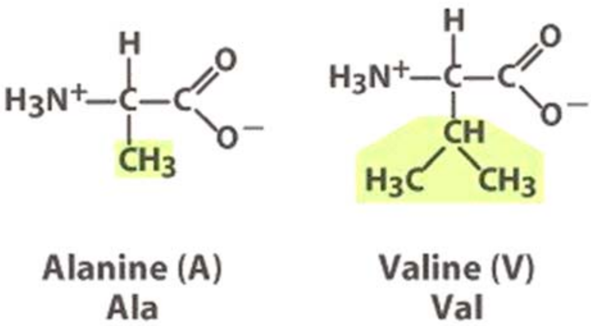


Fig. 3.2

Fig. 3.2 shows the structures of alanine and valine. Both amino acids contains non-polar R group. Suggest how a change from alanine to valine can result in a mutated amyloid precursor protein which then give rise to amyloid plaques. [2]

Total:[7]

For Examiner's use

Question 4

Dengue is the most rapidly spreading mosquito-borne viral disease in the world. In the last 50 years, incidence has increased 30-fold with increasing geographic expansion to new countries and, in the present decade, from urban to rural settings. An estimated 50 million dengue infections occur annually and approximately 2.5 billion people live in dengue endemic countries.

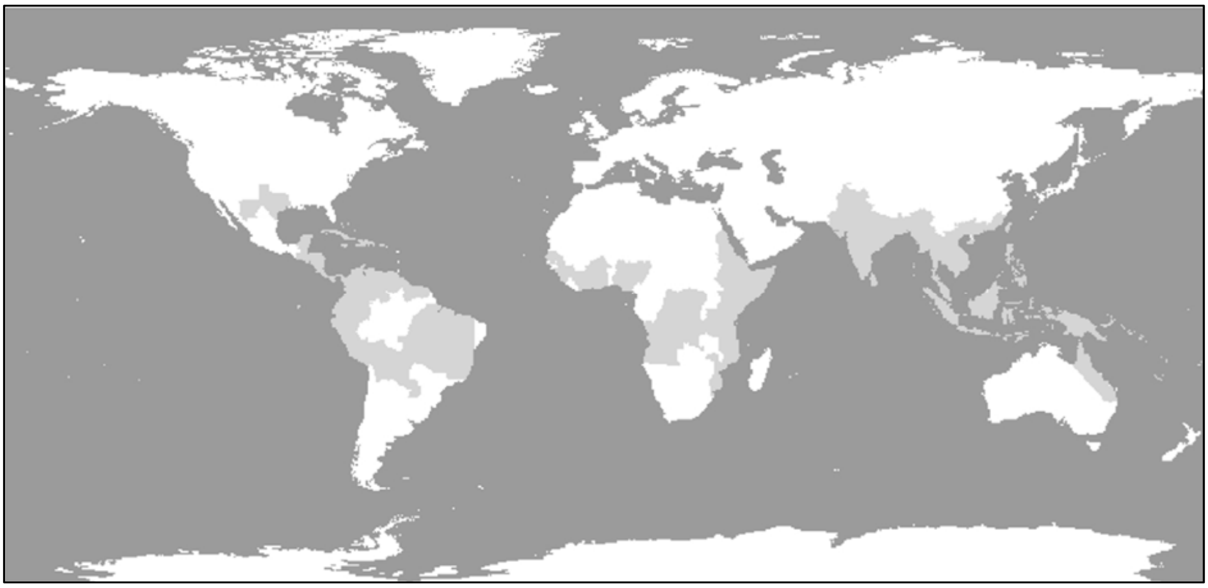


Fig. 4

Shaded areas in Fig. 4 represents countries at risk of dengue fever due to presence of Aedes mosquito, as of 2008.

(a) Outline the developmental stages in the life cycle of the Aedes mosquito. [4]

For Examiner's use

(b) Explain why the growing area of influence of dengue fever overlaps with that of the Aedes mosquito. [3]

(c) To some extent the range of the Aedes mosquito has also followed human expansion. Explain how this may be true. [1]

Total:[8]

Question 5

For Examiner's use

Cranes are large birds. One of the earliest methods of classifying cranes was based on the calls they make during the breeding season.

(a) Suggest why biologists could use calls to investigate relationships between different species of crane. [2]

The bone protein, osteocalcin, has been extracted from fossil Neanderthal skulls found in Shanidar Cave, Iraq. The skulls were estimated to be approximately 75 000 years old.

Using mass spectrometry, the amino acids of this Neanderthal osteocalcin have been sequenced. This sequence has been compared with that in osteocalcin from *Homo sapiens* and other modern primates.

The sequences of the first 20 amino acids in the primary structure of osteocalcin from this study are shown below in Fig. 5. Amino acids are represented by capital letters.

	1 · · · · · · · · 10 · · · · · · · · · 20
Modern human	Y L Y Q W L G A P V P Y P D P L E P R R
Neanderthal	Y L Y Q W L G A P V P Y P D P L E P R R
Chimpanzee	Y L Y Q W L G A P V P Y P D P L E P R R
Orang-utan	Y L Y Q W L G A P V P Y P D P L E P K R
Gorilla	Y L Y Q W L G A O V P Y P D P L E P K R
Monkey	Y L Y Q W L G A O A P Y P D P L E P K R

Fig. 5

(b) Suggest what the data indicate about the relationships between Neanderthals and modern primates. Give reasons for your answer. [4]

Total: [6]

For Examiner's use

Question 6

The following experiment was carried out to investigate the effect of light intensity on the rate of photosynthesis of a water plant, *Elodea*.

- *Elodea* was cut into three pieces, each 10 cm long.
- Each piece of *Elodea* was placed in a glass tube, containing 0.5% sodium hydrogen carbonate solution, which was then sealed with a bung.
- Tube **A** was placed 10 cm away from a lamp.
- Tube **B** was placed 5 cm away from a lamp.
- Tube **C** was placed in a dark room.
- An oxygen sensor was used to measure the percentage of oxygen in the solutions at the start of the experiment and again at 5, 10 and 20 minutes.

The results are shown in Fig. 6.1.

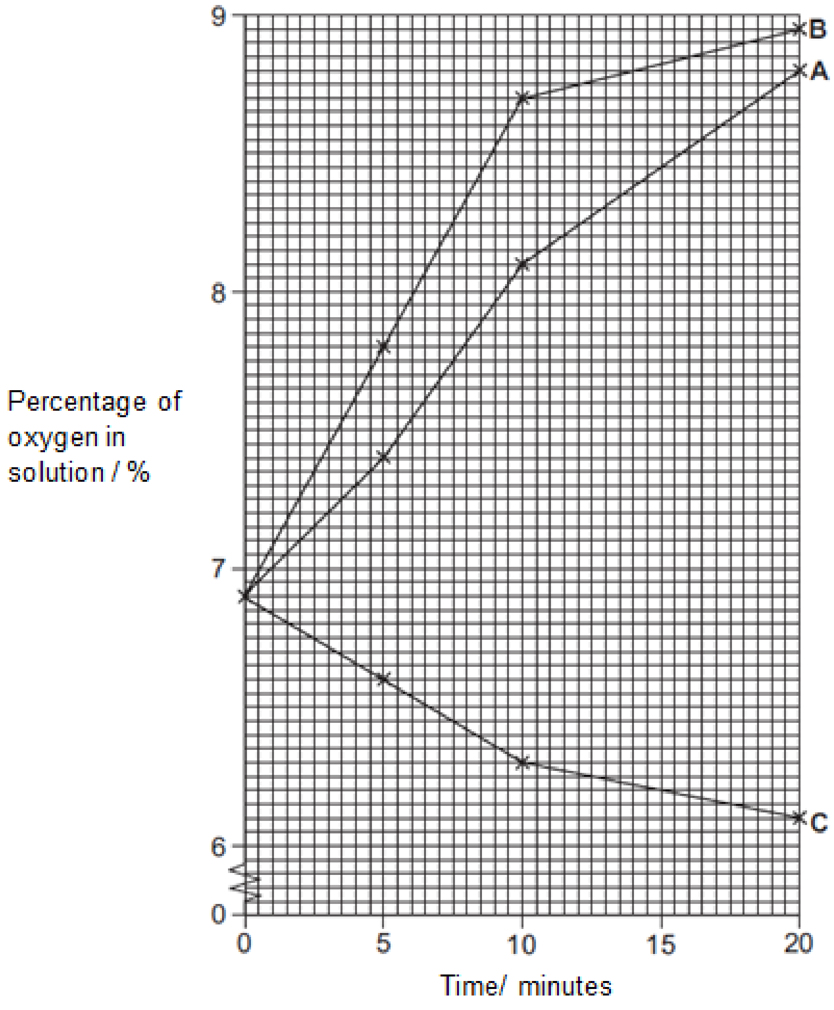


Fig. 6.1

For Examiner's use

(a) With reference to Fig. 6.1,

(i) Calculate the rate of oxygen production for tube **A** for the 20 minutes of the experiment. Give your answer to two decimal places. Show your working. [1]

(ii) Compare the results for tubes **A** and **B**. [2]

(iii) Explain the results for tube **C**. [2]

For Examiner's use

Fig. 6.2 outlines the early stages of respiration in yeast cells in the presence of oxygen.

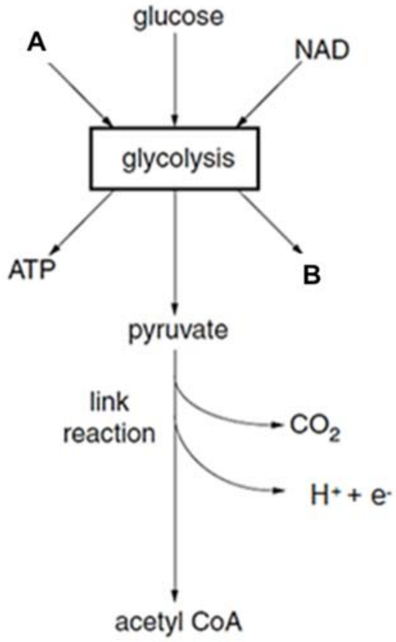


Fig. 6.2

(b) With reference to Fig. 6.2,

(i) Identify molecules **A** and **B**. [2]

A: ....

B: ....

(ii) Explain what would happen to pyruvate in the absence of oxygen. [2]

....

....

....

....

Total: [9]

Section B: Free-Response Question (15 marks)

Answer only **one** question.
Write your answers on the writing paper provided.
Answer each part (a) and (b) on a fresh piece of writing paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.
A **NIL RETURN** is required.

Question 1

(a) Describe the role of proteins in the structure of the cell surface membrane. [5]

(b) Describe ATP synthesis in various cellular processes. [10]

Total: [15]

OR

Question 2

(a) Describe the structure of DNA and explain how its stability is maintained. [5]

(b) Describe how cancer occurs and outline the factors that increase the chances of cancerous growth. [10]

Total: [15]

END OF PAPER

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